

Ved Chauhan

Software Engineer • Full-Stack, Backend & Developer Tools • TypeScript, Rust, Node.js

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SUMMARY

Software engineer who ships production systems end-to-end — relational data models, typed APIs, React frontends, native binaries, and the CI/CD that releases them. Founder of SNOWBROS, building a Rust static-analysis toolchain published on npm and crates.io plus full-stack retail software deployed for a paying client. Strong across TypeScript and Rust, from schema and access control to LSP servers and cross-platform distribution.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, SQL, HTML, CSS | **Frontend:** React 19, Next.js (App Router, SSR/SSG), Tailwind CSS, Radix UI, TanStack Query/Table, Zustand, GSAP

Backend & Data: Node.js, REST APIs, Server Actions, PostgreSQL, SQLite, Drizzle ORM, Supabase, MongoDB, schema design, Row-Level Security, auth, HMAC webhooks, payments (Razorpay UPI), offline sync | **Systems & Tooling:** Tree-sitter, static & semantic analysis, symbol/import graphs, LSP, Tauri, CLI design, SARIF, incremental caching

Testing & DevOps: Vitest, Playwright, Rust integration tests, GitHub Actions (CI/CD), cross-platform release pipelines, npm & crates.io publishing, Docker, Vercel | **Foundations:** Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Founder & Software Engineer — SNOWBROS (Independent Software Studio)

Feb 2026 – Present

TypeScript, Rust, Next.js, Node.js, PostgreSQL, Supabase, Tauri, GitHub Actions

- Own the full lifecycle of shipped software for paying clients and open-source users — requirements, data modeling, implementation, testing, release engineering — with GitHub Actions gating typecheck, lint, unit and E2E runs on every push.
- Built and released two flagship products: **Snowbros Atlas**, a Rust static-analysis toolchain published to npm, crates.io and Homebrew, and **Atelier**, a full-stack retail operating system running in a live boutique.
- Designed multi-tenant PostgreSQL schemas with Row-Level Security, server-side authorization guards, and HMAC-verified payment webhooks so client data and money movement stay isolated and auditable.

PROJECTS

Snowbros Atlas — Open-Source Static Analysis Toolchain — github.com/snowbros-labs/atlas

Jul 2026 – Present

Rust, TypeScript, Tree-sitter, petgraph, tower-lsp, Node.js, GitHub Actions

- Architected a 19-crate Rust workspace (~17k LOC, 340+ tests) that parses JavaScript/TypeScript and Python with Tree-sitter, lowers both into one shared semantic IR, and runs 23 language-agnostic rules over a whole-project import and symbol graph.
- Implemented semantic analysis over the graph — SCC-based cycle detection, dead-file reachability, cross-file call/type-reference edges, Next.js server/client boundary tracking through aliases and re-export chains — each finding carrying an evidence chain and confidence level.
- Engineered an incremental xxh3 + mtime cache with byte-identical warm output: a 500-file repo analyzes in ~270 ms cold, ~34 ms after a one-file change; axios (431 files) in ~230 ms cold / ~76 ms warm. Automated release to npm (@snowbros/atlas), crates.io, Homebrew and x64/arm64 binaries for three platforms across 5 CI workflows.
- Shipped an LSP server plus a TypeScript VS Code extension for live in-editor diagnostics, a multi-command CLI with guarded auto-fixes, and SARIF 2.1.0 output feeding GitHub code scanning behind a --ci gate.

Atelier by Vinny's — Retail Operating System (Client Deployment) — atelier.vinnysvogue.in

Aug 2026 – Present

Next.js 16, React 19, TypeScript, PostgreSQL, Drizzle ORM, Supabase, Tauri, Rust, SQLite, Razorpay, Playwright

- Built and deployed a full-stack business management system for a real fashion boutique — billing with PDF invoice generation, product/variant catalogue, variant-level inventory, customer profiles, staff roles, and revenue reporting derived from settled bills.
- Modeled a 35-table PostgreSQL schema with Drizzle migrations, database-enforced Row-Level Security policies, and a single server-side guard every Server Action passes through, keeping authorization out of the UI layer.
- Integrated Razorpay UPI payments end-to-end: QR checkout, status/cancel/reconciliation APIs, and an HMAC-verified webhook as the sole authority that can settle an invoice — the point-of-sale never holds a payment secret.
- Built an offline-first sync engine for the Tauri desktop till — local SQLite (rusqlite) sales, durable outbox, bidirectional push/pull, conflict resolution, background sync — shipping one codebase to web and a signed Windows build, guarded by 268 Vitest tests and Playwright E2E in CI.

Mentor — Open-Source AI Orchestration Skill — github.com/snowbros-labs/mentor-skill

Jul 2026

Claude Code skills, LLM orchestration, prompt/context engineering, Markdown DSL, Git hooks

- Designed an orchestration layer over ten specialist engineering capability modules that classifies task complexity, routes each task to the minimum capability set, and runs only the review gates the change implicates.
- Used progressive disclosure as the context-budget mechanism — unloaded references cost zero tokens — with checkpoint/resume state so long tasks survive context limits; released open-source under MIT.

STC Academy — Multi-Role Learning Management System — stc.vercel.app

Mar 2026 – Jul 2026

Next.js, React, Node.js, TypeScript, PostgreSQL, Supabase

- Shipped a production portal for a 100+ user academy as sole engineer, replacing paper logs and spreadsheets with role-restricted dashboards for students, teachers and administrators over secured API routes.
- Designed a normalized relational schema with database triggers automating attendance tallies and Row-Level Security enforcing per-student privacy at the data layer.

Vinnys Vogue — Full-Stack E-Commerce Platform — vinnysvogue.in

Feb 2026

Next.js, React, Node.js, TypeScript, Supabase, Tailwind CSS

- Delivered a production storefront as sole engineer — catalogue, cart, checkout, inventory tracking, admin dashboard and weekly sales exports — from schema design to deployment, keeping stock consistent under concurrent checkout.

VED.EXE — Interactive Developer Portfolio — ved.exe.snowbros.me

Jun 2026 – Jul 2026

Next.js, React, TypeScript, GitHub REST API, GSAP, Framer Motion

- Built a terminal-inspired portfolio with a command parser, live GitHub REST API contribution metrics, and performant responsive animations.

EDUCATION

B.Tech in Computer Engineering — U.V. Patel College of Engineering

Expected May 2028

CGPA: 7.5 / 10.0 | Mehsana, Gujarat, India | IEEE Student Member (Jul 2024 – Present)